05 Is AI reorienting archaeological methods?

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Generative AI and the practice of 'vibe coding' are changing how archaeologists carry out computational research, but their effects on methodological diversity remain unclear. In this paper, we investigate whether large language models (LLMs) are contributing to a narrowing of the methods used in archaeology.

We combine an analysis of archaeological literature with a controlled experiment on LLM-based method recommendations. First, we analysed ~119,000 archaeology abstracts published between 2010 and 2025, using a locally run LLM to identify and classify the computational methods reported in the literature. We then modelled changes in method composition across archaeological sub-disciplines, focusing on whether method use changed following the emergence of generative AI. Although we find a small but credible shift after 2023, this change is smaller than the variation observed across the wider study period. No individual technique shows a significant increase or decrease, and overall methodological diversity has increased rather than declined.

Second, we asked two open-weight LLMs to recommend computational methods for 28 archaeological research problems, using prompts with different levels of methodological guidance. The methods recommended by the models were substantially less diverse than those found in the published literature, particularly when little methodological guidance was provided. The models also tended to favour established techniques and produced recommendation profiles more similar to the post-2023 literature.

Our findings do not demonstrate that LLMs are causing methodological convergence, but they suggest that the ways in which archaeologists interact with these systems may favour a narrower set of established approaches. We discuss what this might mean for methodological diversity in computational archaeology as LLMs become increasingly embedded in research practice.